

Ishaan Chatterjee

Embedded Systems & IoT Firmware Engineer • 6 Years Experience • MERIQ Index: 70% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: MQTT, C/C++, PCB Debugging, FreeRTOS, Python, Embedded Linux, ARM Cortex

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Tesla* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core MQTT service, reducing latency by 59%.
- Led a squad of 5 engineers delivering scalable microservices processing 33M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 69%.

Associate Embedded Systems & IoT Firmware Engineer — *Intel* (2020 - 2023)

- Designed and maintained C/C++ data endpoints with 97% SLA uptime.
- Optimized database indexing and caching layers using PCB Debugging saving \$14k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [MQTT, C/C++, PCB Debugging, FreeRTOS]

Engineered end-to-end production solution leveraging MQTT and C/C++ with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [PCB Debugging, FreeRTOS, Python, Embedded Linux]

Architected high-resilience system integrating PCB Debugging with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Electrical and Computer Engineering

National University of Singapore — 2016 - 2020 (Score: 3.7 / 4.0)